

Random Variables and Random Number Generation

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Abstract

This report explores random number generation and density estimation techniques. We compare histogram and kernel density estimation (KDE) methods, demonstrating their trade-offs in accuracy and smoothness. Probability transformations using the Jacobian formula are applied to derive distributions for linear and quadratic functions of normal random variables. The inverse CDF method is implemented to generate exponential random variates. Finally, the alpha-stable distribution is explored, demonstrating heavy-tailed random variable generation with varying stability and skewness parameters.

1 Uniform and Normal Random Variables

1.1 Kernel Density Estimation

The kernel density estimator is defined as

$$\hat{p}(x) = \frac{1}{N} \sum_{i=1}^N \frac{1}{\sigma} \mathcal{K}\left(\frac{x - x^{(i)}}{\sigma}\right) \quad (1)$$

where N is the number of samples, $x^{(i)}$ are the observed data points, $\mathcal{K}(u)$ is the kernel function, and σ is the bandwidth. The Gaussian kernel is:

$$\mathcal{K}_{\text{Gauss}}(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) \quad (2)$$

resulting in the KDE:

$$\hat{p}(x) = \frac{1}{N\sigma} \sum_{i=1}^N \exp\left(-\frac{(x - x^{(i)})^2}{2\sigma^2}\right) \quad (3)$$

1.2 Histogram and KDE of Gaussian Random Numbers

The histogram (blue bars) and KDE (blue curve) both capture the bell-shaped Gaussian profile. The histogram exhibits discontinuities at bin boundaries while the KDE produces a smooth estimate that closely follows the theoretical PDF (red curve).

1.3 Histogram and KDE of Uniform Random Numbers

The histogram shows approximately equal heights across bins, consistent with uniform density. The KDE ($\sigma = 0.05$) is smoother but extends beyond $[0, 1]$ due to boundary effects.

1.4 Comparison of Methods

Both methods have similar bias-variance structure but differ in asymptotic convergence. From statistical theory [1]:

Histogram: Optimal convergence rate $O(N^{-2/3})$

KDE: Optimal convergence rate $O(N^{-4/5})$ (faster)

Table 1: Comparison of histogram and KDE methods.

Aspect	Histogram	KDE
Visual Form	Discrete bars	Smooth curve
Interpretation	Bin counts	Smoothed est.
Param. Sens.	High	Lower
Boundary	Respects	May exceed
Convergence	$O(N^{-2/3})$	$O(N^{-4/5})$
Comp. Cost	Trivial	$O(N)$

Histogram advantages: Directly represents sample counts per bin (intuitive), respects finite support of distributions, computationally simple.

Histogram disadvantages: Sensitive to bin width and position, artificial discontinuities between bins.

KDE advantages: Smooth, continuous estimate revealing density structure, superior convergence rate, less sensitive to bandwidth selection.

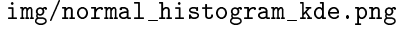
KDE disadvantages: Less intuitive, boundary effects for finite support distributions, computationally more expensive.

1.5 Theoretical Analysis of Uniform Density

When histogramming N samples from $\mathcal{U}(0, 1)$ into J bins, each bin has width $\delta = 1/J$ and probability $p_j = 1/J$. Each marginal count follows:

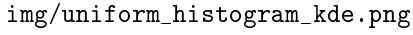
$$n_j \sim \text{Binomial}\left(N, p_j = \frac{1}{J}\right) \quad (4)$$

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img/normal_histogram_kde.png

Figure 1: Normal distribution: histogram (left) and KDE (right) compared with theoretical $\mathcal{N}(0,1)$ PDF. The histogram exhibits discontinuities at bin boundaries while KDE ($\sigma = 0.2$) produces a smooth estimate.



img/uniform_histogram_kde.png

Figure 2: Uniform distribution: histogram (left) and KDE (right) compared with theoretical $\mathcal{U}(0,1)$ PDF. KDE extends beyond $[0,1]$ due to boundary effects.

Therefore:

$$E[n_j] = \frac{N}{J} \quad (5)$$

$$\text{Var}(n_j) = \frac{N(J-1)}{J^2} \quad (6)$$

$$\sigma_{n_j} = \frac{\sqrt{N(J-1)}}{J} \quad (7)$$

The coefficient of variation is $\text{CV} = \sqrt{(J-1)/N} \approx \sqrt{J/N}$. As $N \rightarrow \infty$, $\text{CV} \rightarrow 0$, so histogram estimates become increasingly reliable.

The histogram results demonstrate good agreement with theory:

1. **Bin Count Distribution:** Expected mean $E[n_j] = N/J$ closely matches observed counts across all N values.
2. **Confidence Bounds:** Approximately 95–100% of bins fall within $\pm 3\sigma$ bounds, consistent with the $\sim 99.7\%$ theoretical expectation from CLT.
3. **Increasing Reliability:** As N increases, relative scatter decreases proportionally to $1/\sqrt{N}$.

2 Functions of Random Variables

2.1 Linear Transformation

For $x \sim \mathcal{N}(0,1)$ and $y = ax + b$, the inverse is $x = (y - b)/a$ with Jacobian $|dx/dy| = 1/|a|$. Applying the transformation formula:

$$\begin{aligned} p(y) &= p(x) \left| \frac{dx}{dy} \right| \\ &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(y-b)^2}{2a^2}\right) \cdot \frac{1}{|a|} \\ &= \frac{1}{\sqrt{2\pi}a^2} \exp\left(-\frac{(y-b)^2}{2a^2}\right) \\ &= \mathcal{N}(y|b, a^2) \end{aligned} \quad (8)$$

This demonstrates that a linear transformation of standard normal produces $\mathcal{N}(\mu, \sigma^2)$ with $\mu = b$ and $\sigma^2 = a^2$. Any normal distribution can be generated from $\mathcal{N}(0,1)$ using $Y = \sigma X + \mu$.

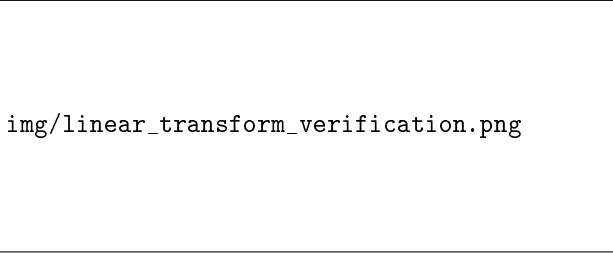
2.2 Quadratic Transformation

For $x \sim \mathcal{N}(0,1)$ and $y = x^2$ ($y \geq 0$), there are two inverses: $x_1(y) = \sqrt{y}$ and $x_2(y) = -\sqrt{y}$. The Jacobian



img/multinomial_bounds.png

Figure 3: Histograms for $N = 100, 1000, 10000$ with theoretical mean (red) and $\pm 3\sigma$ bounds (green). As N increases, relative scatter decreases proportionally to $1/\sqrt{N}$.



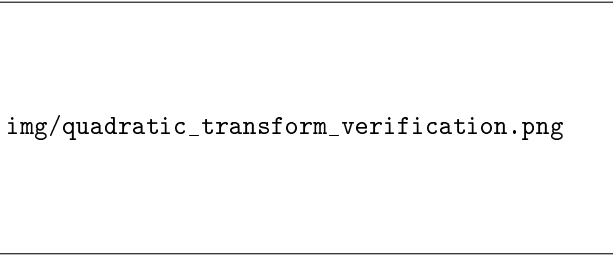
img/linear_transform_verification.png

Figure 4: Verification of linear transformation $y = 3x + 5$ where $x \sim \mathcal{N}(0, 1)$. The histogram matches theoretical $\mathcal{N}(5, 9)$.

is $|dx/dy| = 1/(2\sqrt{y})$. Applying the formula:

$$\begin{aligned} p(y) &= 2 \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y}{2}\right) \cdot \frac{1}{2\sqrt{y}} \\ &= \frac{1}{\sqrt{2\pi y}} \exp\left(-\frac{y}{2}\right), \quad y \geq 0 \end{aligned} \quad (9)$$

This is the chi-squared distribution with 1 degree of freedom, χ_1^2 .



img/quadratic_transform_verification.png

Figure 5: Verification of quadratic transformation $y = x^2$. Sample mean ≈ 1.0 and variance ≈ 2.0 match theoretical χ_1^2 moments.

3 Inverse CDF Method

3.1 Theory

If $U \sim \mathcal{U}(0, 1)$ and $Y = F^{-1}(U)$ where F is any CDF, then Y has distribution with CDF F . The algorithm:

1. Generate $u^{(i)} \sim \mathcal{U}(0, 1)$
2. Compute $y^{(i)} = F^{-1}(u^{(i)})$

3.2 Exponential Distribution

For the exponential distribution with PDF $p(y) = \exp(-y)$ for $y \geq 0$:

CDF:

$$F(y) = \int_0^y \exp(-t) dt = 1 - \exp(-y) \quad (10)$$

Inverse CDF: Solving $u = F(y)$ for y :

$$u = 1 - \exp(-y) \implies y = -\ln(1 - u) = F^{-1}(u) \quad (11)$$

Since $(1 - u) \sim \mathcal{U}(0, 1)$ when $u \sim \mathcal{U}(0, 1)$, we can use $F^{-1}(u) = -\ln(u)$.

3.3 Implementation

```
def generate_exponential(n, mean=1.0):
    u = np.random.rand(n)
    return -mean * np.log(u)
```

The histogram shows excellent agreement with the theoretical exponential density. The KDE exhibits boundary effects near $y = 0$ where Gaussian kernels extend beyond the valid support $[0, \infty)$.

img/exponential_inverse_cdf.png

Figure 6: Exponential distribution via inverse CDF method. Histogram (left) matches theoretical PDF. KDE (right) shows boundary effects: peak underestimated (≈ 0.65 vs 1.0) due to Gaussian kernels extending into $y < 0$.

4 Alpha-Stable Distribution

4.1 Generation Algorithm

The α -stable distribution models heavy-tailed phenomena in communications and signal processing. To generate samples with $\alpha \in (0, 2)$, $\alpha \neq 1$ and $\beta \in [-1, +1]$:

1. Calculate auxiliary parameters:

$$b = \frac{1}{\alpha} \arctan\left(\beta \tan \frac{\pi\alpha}{2}\right) \quad (12)$$

$$s = \left(1 + \beta^2 \tan^2 \frac{\pi\alpha}{2}\right)^{1/(2\alpha)} \quad (13)$$

2. Generate $U \sim \mathcal{U}(-\pi/2, +\pi/2)$ and $V \sim \mathcal{E}(1)$
3. Calculate:

$$X = s \frac{\sin(\alpha(U+b))}{(\cos U)^{1/\alpha}} \left(\frac{\cos(U - \alpha(U+b))}{V} \right)^{\frac{1-\alpha}{\alpha}} \quad (14)$$

4.2 Implementation

```
def generate_stable(alpha, beta, n):
    b = (1/alpha) * np.arctan(
        beta * np.tan(np.pi*alpha/2))
    s = (1 + beta**2 * np.tan(
        np.pi*alpha/2)**2)**(1/(2*alpha))
    U = np.random.uniform(
        -np.pi/2, np.pi/2, n)
    V = np.random.exponential(1.0, n)
    num = np.sin(alpha*(U+b)) / \
        np.cos(U)**(1/alpha)
    den = (np.cos(U-alpha*(U+b))/V) \
        **((1-alpha)/alpha)
    return s * num * den
```

4.3 Parameter Interpretation

Stability parameter $\alpha \in (0, 2)$: Controls tail heaviness. At $\alpha = 0.5$, tails are extremely heavy with massive outliers. At $\alpha = 1.5$, tails are lighter but still heavier than Gaussian. As $\alpha \rightarrow 2$, the distribution approaches Gaussian.

Skewness parameter $\beta \in [-1, +1]$: Controls asymmetry. At $\beta = 0$, the distribution is symmetric. Positive β produces right skewness; negative β produces left skewness.

5 Conclusion

This report explored fundamental random number generation and density estimation techniques. Histogram and KDE methods offer complementary trade-offs: histograms are intuitive and respect bounded support, while KDE provides smoother estimates with faster convergence. Jacobian transformations enable derivation of transformed variable distributions, as demonstrated for linear ($y = ax + b \Rightarrow \mathcal{N}(b, a^2)$) and quadratic ($y = x^2 \Rightarrow \chi_1^2$) cases. The inverse CDF method provides a general framework for generating samples from any distribution with invertible CDF. Finally, alpha-stable distributions demonstrate flexible modeling of heavy-tailed phenomena beyond Gaussian assumptions.

References

- [1] University of Washington. *STAT 425: Course Lecture Notes on Density Estimation*. Lecture notes covering histogram and kernel density estimation methods. 2018. URL: https://faculty.washington.edu/yenchic/18W_425/Lec6_hist_KDE.pdf.

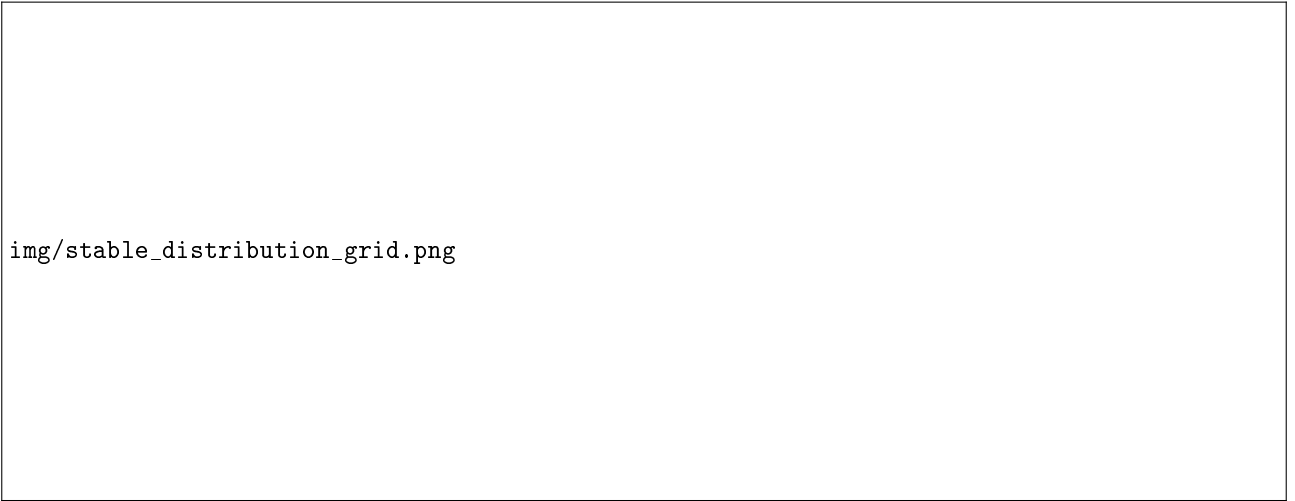


Figure 7: Alpha-stable distributions for $\alpha = 0.5$ (top, heavy tails) and $\alpha = 1.5$ (bottom, lighter tails) with varying β (skewness). Heavy-tailed distributions ($\alpha = 0.5$) produce extreme outliers requiring plot clipping.